

Kuiyi (Cory) Gao

Chapel Hill, NC kuiyigao@unc.edu kuiyigao.github.io GitHub

Languages

Mandarin (native), English (TOEFL iBT 110), Cantonese (intermediate).

Research Focus

AI safety, agent reliability, multimodal safety evaluation, runtime monitoring, and human oversight.

Honors and Awards

Professor Joseph Lau Scholarship of Innovation

CUHK Faculty of Engineering ELITE Stream Award

CUHK College Entrance Scholarship CUHK Exchange Scholarship, UNC Spring 2026

Education

University of North Carolina at Chapel Hill

Undergraduate, intended B.A. in Computer Science

GPA 3.68/4.00. Coursework in algorithms, machine learning, natural language processing, technology ethics, C and Unix systems, and Java object-oriented design.

Jan 2026 to present

Expected May 2027

The Chinese University of Hong Kong

Undergraduate study in Computer Science, transferred to UNC

Graduate-level deep learning, artificial intelligence, data structures, formal languages, computer organization, probability, statistics, and linear algebra.

Sep 2023 to May 2026

Publications and Manuscripts

[1] **Kuiyi Gao*** and collaborators. *RealGUINoise: A Dynamic Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. Co-first author. Under review, 2026.

[2] **Kuiyi Gao**. *RuntimeTickets: Enforcing and Auditing Task-Specific Security Requirements for Third-Party Agent Skills*. First author. Under review at ICLR 2027.

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 Main Conference. [paper] [code]

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [paper] [code]

Research Experience

RealGUINoise [1]

CUHK, advisor: Prof. Michael R. Lyu

Co-first author. Built paired web and Android tasks with controlled interface noise, designed the evaluation methodology, implemented the automated judging pipeline, and ran clean and altered-interface experiments. Measured task completion, unsafe actions, and action-sequence differences.

Hong Kong and remote

Sep 2025 to present

RuntimeTickets [2] [code]

MBZUAI UGRIP and continuing research, advisor: Dr. Steve Liu

First author. Designed task-specific contracts that narrow host permissions, gate runtime requests, and retain decision and effect records for later audit. Built the runtime firewall in a four-student UGRIP team and lead method design, experiment engineering, and manuscript development.

Abu Dhabi and Beijing

Jun 2026 to present

Research Experience (continued)

Chain-of-Jailbreak [4]

Tencent AI Lab and CUHK

Second author. Designed a multi-turn image-editing jailbreak method and prompting-based defense, contributed benchmark data, and ran most attack and defense experiments. Built evaluation scripts, prepared paper figures, and contributed to the manuscript and rebuttal.

Hong Kong

Jun 2024 to Feb 2025

MMSafeAware [3]

Tencent AI Lab

Third author. Contributed benchmark construction and evaluation design for 1,500 image-text pairs across 29 safety scenarios. Ran vision-centric fine-tuning experiments, separated missed unsafe inputs from false alarms on benign inputs, and contributed to submission and rebuttal.

Hong Kong and Shenzhen

Dec 2023 to Oct 2024

Selected Projects

MemSlot: Long-Document Compression [code]

2026

Natural language processing course project

Tests whether learned word saliency can reduce contract text before OCR and question answering while retaining relevant content. Implemented memory-slot attention over frozen RoBERTa features, overlap-aware word scoring, saliency-guided rendering, and text-pruning baselines, while the full OCR to QA evaluation remains incomplete.

Poetry and Image Variation

2025

AI and language course project

Examines whether poem style and prompt length relate to variation among generated images. Analyzed saved CLIP-based variation scores for Imagist and Romantic excerpts generated with FLUX and Stable Diffusion using Python, SciPy, and Matplotlib.

Text-KEN: Kernel Comparisons for Sentence Embeddings

May to Aug 2025

CUHK summer research, advisor: Prof. Farzan Farnia

Tests a nonparametric way to compare two text collections without training a classifier. Built a Python prototype using MiniLM sentence embeddings, RBF kernels, spectral entropy, centering, and covariance whitening.

Movability Segmentation [code]

2025

Deep learning course project

Maps scene categories to four movability classes for an interpretable semantic-segmentation task. Implemented DeepLabV3 with a custom ASPP head, COCO-Stuff label remapping, image-mask preprocessing, and checkpoint-based inference.

Technical Skills

Research engineering Python, Git, Linux, Docker, Playwright, HTTP and JSON APIs, experiment configuration, trace logging.

Machine learning PyTorch, Hugging Face Transformers, sentence-transformers, attention modules, frozen encoders, fine-tuning, semantic segmentation.

Evaluation and analysis Benchmark construction, task success and action metrics, multimodal judgment, span-QA exact match and F1, kernel methods, statistical testing.

Data and visualization NumPy, pandas, SciPy, Matplotlib, seaborn, Pillow, document rendering, image grids.

Course foundation Algorithms, machine learning, natural language processing, deep learning, computer vision, artificial intelligence, probability, linear algebra.